RED-diff++

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Abstract

1 Introduction

2 "Theory"

RED-diff is based on the following variational approximation. We wish to sample from $p(\mathbf{x}_0 \mid \mathbf{y})$ by fitting a Gaussian variational distribution $q = \mathcal{N}(\boldsymbol{\mu}, \sigma^2\mathbf{I})$. This is essentially solving the MAP estimation problem. We attempt to minimize the KL divergence between q and p, which can be written as:

$$D_{\mathrm{KL}}(q(\mathbf{x}_0 \mid \mathbf{y}) \parallel p(\mathbf{x}_0 \mid \mathbf{y})) = -\mathbb{E}_{q(\mathbf{x}_0|\mathbf{y})}[\log p(\mathbf{y} \mid \mathbf{x}_0)] + D_{\mathrm{KL}}(q(\mathbf{x}_0 \mid \mathbf{y}) \parallel p(\mathbf{x}_0)) + \log p(\mathbf{y}). \tag{1}$$

The final term $\log p(\mathbf{y})$ is a constant with respect to the parameters of q and can thus be ignored. Assuming that the forward model is Gaussian, the first term can be rewritten as:

$$-\mathbb{E}_{q(\mathbf{x}_0|\mathbf{y})}[\log p(\mathbf{y} \mid \mathbf{x}_0)] = \frac{-1}{2\sigma_{\text{obs}}^2}\mathbb{E}_{q(\mathbf{x}_0|\mathbf{y})}\left[||\mathcal{A}(\mathbf{x}) - \mathbf{y}||_2^2\right]. \tag{2}$$

To deal with the middle term, I think RED-diff uses the following lemmas from [1], although it is not really clear. The first result is:

$$\frac{\partial D_{\mathrm{KL}}(p_t \parallel q_t)}{\partial t} = -\frac{1}{2}\int p_t(\mathbf{x})g(t)^2||\nabla_{\mathbf{x}}\log p_t(\mathbf{x}) - \nabla_{\mathbf{x}}\log q_t(\mathbf{x})||_2^2\, d\mathbf{x}, \tag{3}$$

where $g(t)$ is the diffusion coefficient (so for EDM $g(t) = \sigma(t)$). According to [1], this implies that

$$D_{\mathrm{KL}}(p_t \parallel q_t) = -\frac{1}{2}\int_0^1 \mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})}\left[g(t)^2||\nabla_{\mathbf{x}}\log p_t(\mathbf{x}) - \nabla_{\mathbf{x}}\log q_t(\mathbf{x})||_2^2\right] dt. \tag{4}$$

In RED-diff, they use this to claim that:

$$D_{\mathrm{KL}}(q(\mathbf{x}_0 \mid \mathbf{y}) \parallel p(\mathbf{x}_0)) = \int_0^1 \frac{\beta(t)}{2}\mathbb{E}_{\mathbf{x}\sim p_t(\mathbf{x})}\left[||\nabla_{\mathbf{x}}\log q(\mathbf{x}_t \mid \mathbf{y}) - \nabla_{\mathbf{x}}\log p_t(\mathbf{x})||_2^2\right] dt. \tag{5}$$

(We will need to convert between β and g, since β comes from the VP-SDE.) Conveniently, $\log q(\mathbf{x}_t \mid \mathbf{y})$ is available in closed form since q is Gaussian. RED-diff adds a weighting term in an ad hoc manner. I think we can more rigorously justify this using the theory in [2].

3 Results

Method	360 receivers		180 receivers		60 receivers	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
DPS†	32.061 (2.163)	0.846 (0.127)	31.798 (2.163)	0.862 (0.123)	27.372 (3.415)	0.813 (0.133)
PnP-DM†	33.914 (2.054)	0.988 (0.006)	31.817 (2.073)	0.981 (0.008)	24.715 (2.874)	0.909 (0.046)
DAPS†	34.641 (1.693)	0.957 (0.006)	33.160 (1.704)	0.944 (0.009)	25.875 (3.110)	0.885 (0.030)
RED-diff	36.569 (2.294)	0.980 (0.004)	35.412 (2.148)	0.984 (0.004)	27.103 (3.316)	0.935 (0.036)
RED-diff++	39.043 (1.898)	0.985 (0.003)	37.877 (1.819)	0.987 (0.002)	30.412 (3.547)	0.966 (0.022)

Table 1: Quantitative comparison of algorithms for linear inverse scattering. Noise level $\sigma_y = 10^{-4}$ †Indicates that we use the result reported in [3].

Subsampling ratio	×4						×8					
Measurement type	Simulated (noiseless)			Raw			Simulated (noiseless)			Raw		
Methods	PSNR↑	SSIM↑	Data misfit↓	PSNR↑	SSIM↑	Data misfit↓	PSNR↑	SSIM↑	Data misfit↓	PSNR↑	SSIM↑	Data misfit↓
DiffPIR†	28.31 (1.598)	0.632 (0.107)	10.545 (2.466)	27.60 (1.470)	0.624 (0.111)	34.015 (15.522)	26.78 (1.556)	0.588 (0.113)	7.787 (1.741)	26.26 (1.458)	0.590 (0.113)	24.208 (10.922)
DPS†	26.13 (4.247)	0.620 (0.105)	9.900 (2.925)	25.83 (2.197)	0.548 (0.116)	35.095 (15.967)	20.82 (4.777)	0.536 (0.111)	6.737 (1.928)	23.00 (3.205)	0.507 (0.109)	24.842 (11.263)
DAPS†	31.48 (1.988)	0.762 (0.089)	1.566 (0.390)	28.61 (2.197)	0.689 (0.102)	31.115 (15.497)	29.01 (1.712)	0.681 (0.098)	1.280 (0.301)	27.10 (2.034)	0.629 (0.107)	22.729 (10.926)
PnP-DM†	31.80 (3.473)	0.780 (0.096)	4.701 (0.675)	27.62 (3.425)	0.679 (0.117)	32.261 (15.169)	29.33 (3.081)	0.704 (0.105)	3.421 (0.504)	25.28 (3.102)	0.607 (0.117)	22.879 (10.712)
RED-diff	29.415 (7.808)	0.732 (0.132)	0.509 (0.076)	28.714 (2.755)	0.626 (0.126)	31.591 (15.368)	26.690 (6.738)	0.646 (0.123)	0.485 (0.067)	27.326 (2.441)	0.563 (0.117)	22.336 (10.838)
RED-diff++	32.334 (5.785)	0.819 (0.112)	0.393 (0.053)	28.540 (3.716)	0.619 (0.134)	31.852 (15.352)	29.878 (4.197)	0.652 (0.124)	0.581 (0.082)	27.428 (1.658)	0.523 (0.135)	23.446 (11.043)

Table 2: Results on compressed sensing MRI (knee). Mean and standard deviation are reported over the test cases. † indicates that we use the result reported in [3].

Observation time ratio	3%				10%				100%			
Methods	PSNR	Blur PSNR	$\tilde{\chi}^2_{cp}$	$\tilde{\chi}^2_{\text{logca}}$	PSNR	Blur PSNR	$\tilde{\chi}^2_{cp}$	$\tilde{\chi}^2_{\text{logca}}$	PSNR	Blur PSNR	$\tilde{\chi}^2_{cp}$	$\tilde{\chi}^2_{\text{logca}}$
DiffPIR†	24.12 (3.25)	30.45 (4.88)	14.085 (14.105)	10.545 (8.860)	23.84 (3.59)	30.04 (5.03)	5.374 (3.733)	5.205 (5.556)	25.01 (4.64)	31.86 (6.56)	3.271 (1.623)	2.970 (1.202)
DPS†	24.20 (3.72)	30.83 (5.58)	8.024 (24.336)	5.007 (5.750)	24.36 (3.72)	30.79 (5.75)	13.052 (43.087)	6.614 (26.789)	25.86 (3.90)	32.94 (6.19)	8.759 (37.784)	5.456 (24.185)
DAPS†	23.54 (3.28)	29.48 (4.88)	3.647 (3.287)	2.329 (1.354)	23.99 (3.56)	30.10 (5.13)	1.545 (0.705)	2.253 (9.903)	25.60 (3.64)	32.78 (5.68)	1.300 (0.324)	1.229 (0.532)
PnP-DM†	24.25 (3.45)	30.49 (4.93)	2.201 (1.352)	1.668 (0.551)	24.57 (3.47)	30.80 (5.22)	1.433 (0.417)	1.336 (0.478)	26.07 (3.70)	32.88 (6.02)	1.311 (0.195)	1.199 (0.221)
RED-diff	20.74 (2.62)	26.10 (3.35)	6.713 (6.925)	9.128 (19.052)	22.53 (3.02)	27.67 (4.53)	2.488 (2.925)	4.916 (13.221)	23.77 (4.13)	29.13 (6.22)	1.853 (0.938)	2.050 (2.361)
RED-diff++	22.027 (3.63)	26.389 (4.97)	1.547 (0.658)	1.689 (0.86)	23.03 (4.089)	27.90 (5.968)	2.54 (1.62)	5.46 (15.67)	24.56 (4.534)	30.436 (7.171)	2.27 (1.429)	3.35 (4.537)

Table 3: Results on black hole imaging. PSNR and Chi-squared of different algorithms on black hole imaging. Gain and phase noise and thermal noise are added based on EHT library. † indicates that we use the result reported in [3].

Method	PSNR↑	SSIM↑	LPIPS↓
DiffPIR	19.140 (3.067)	0.748 (0.079)	0.200 (0.045)
DPS	19.809 (3.209)	0.740 (0.084)	0.206 (0.048)
DAPS	20.397 (3.031)	0.769 (0.073)	0.169 (0.038)
PnP-DM	18.623 (2.842)	0.762 (0.069)	0.207 (0.045)
RED-diff	19.810 (3.025)	0.739 (0.087)	0.196 (0.044)
RED-diff++	20.645 (3.264)	0.835 (0.052)	0.165 (0.034)

Table 4: Quantitative results on FFHQ inpainting.

Method	Gaussian Deblur			Motion Deblur		
	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
DAPS	27.631 (3.237)	0.847 (0.068)	0.143 (0.039)	26.640 (2.877)	0.799 (0.077)	0.157 (0.038)
DPS	25.934 (3.285)	0.798 (0.083)	0.126 (0.033)	25.816 (3.100)	0.792 (0.082)	0.139 (0.032)
PnP-DM	26.022 (3.480)	0.821 (0.080)	0.163 (0.045)	26.187 (3.398)	0.824 (0.077)	0.152 (0.041)
DiffPIR	26.120 (3.118)	0.806 (0.081)	0.133 (0.033)	25.027 (3.063)	0.777 (0.089)	0.161 (0.036)
RED-diff	26.972 (2.820)	0.810 (0.071)	0.220 (0.047)	25.437 (3.240)	0.808 (0.085)	0.212 (0.052)
RED-diff++	27.920 (3.325)	0.858 (0.065)	0.167 (0.043)	27.793 (3.128)	0.853 (0.067)	0.172 (0.044)

Table 5: Quantitative results on FFHQ deblurring.

Method	SR $\times 4$			SR $\times 8$		
	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
DAPS	27.333 (2.999)	0.832 (0.068)	0.141 (0.037)	24.253 (3.078)	0.762 (0.094)	0.224 (0.052)
DPS	26.437 (3.059)	0.809 (0.076)	0.124 (0.031)	22.990 (3.180)	0.728 (0.106)	0.200 (0.044)
PnP-DM	26.511 (3.097)	0.820 (0.071)	0.171 (0.044)	22.328 (3.087)	0.740 (0.107)	0.272 (0.061)
DiffPIR	25.523 (2.646)	0.784 (0.077)	0.257 (0.047)	22.523 (2.694)	0.729 (0.099)	0.379 (0.059)
RED-diff	27.905 (3.214)	0.857 (0.064)	0.161 (0.040)	24.560 (3.165)	0.787 (0.091)	0.256 (0.057)
RED-diff++	27.941 (3.146)	0.860 (0.061)	0.157 (0.040)	24.789 (3.171)	0.792 (0.089)	0.255 (0.055)

Table 6: Quantitative results on FFHQ super-resolution.

Generalization	Vertical $\rightarrow$ Horizontal			Brain $\rightarrow$ Knee			$\times 4 \rightarrow \times 8$		
Methods	PSNR↑	SSIM↑	Data misfit↓	PSNR↑	SSIM↑	Data misfit↓	PSNR↑	SSIM↑	Data misfit↓
DiffPIR†	27.930 (1.502)	0.637 (0.113)	34.188 (15.479)	27.181 (1.496)	0.639 (0.099)	32.542 (15.289)	26.260 (1.458)	0.590 (0.113)	24.208 (10.922)
DPS†	26.770 (1.546)	0.571 (0.117)	35.233 (16.006)	25.432 (2.527)	0.571 (0.123)	33.880 (15.774)	23.000 (3.205)	0.507 (0.109)	24.842 (11.263)
DAPS†	28.780 (2.209)	0.696 (0.105)	32.198 (15.538)	27.268 (2.915)	0.672 (0.123)	31.577 (15.415)	27.100 (2.034)	0.629 (0.107)	22.729 (10.926)
PnP-DM†	27.930 (3.444)	0.689 (0.121)	32.391 (15.235)	27.413 (2.264)	0.640 (0.121)	32.443 (15.235)	25.280 (3.102)	0.607 (0.117)	22.879 (10.712)
RED-diff	28.861 (2.625)	0.628 (0.127)	31.740 (15.421)	28.501 (1.806)	0.600 (0.137)	31.658 (15.384)	27.326 (2.441)	0.563 (0.117)	22.336 (10.838)
RED-diff++	28.878 (3.229)	0.615 (0.135)	31.992 (15.414)	28.758 (2.075)	0.598 (0.130)	31.981 (15.376)	27.610 (3.071)	0.580 (0.135)	22.400 (10.800)

Table 7: Generalization results on compressed sensing MRI with $\times 4$ acceleration and raw measurements. Mean and standard deviation are reported over 96 test cases. † indicates that we use the result reported in [3].

References

[1] Y. Song, C. Durkan, I. Murray, and S. Ermon, "Maximum likelihood training of score-based diffusion models," *Advances in neural information processing systems*, vol. 34, pp. 1415–1428, 2021.

[2] J. Shi and M. K. Titsias, "Demystifying diffusion objectives: Reweighted losses are better variational bounds," *arXiv preprint arXiv:2511.19664*, 2025.

[3] H. Zheng, W. Chu, B. Zhang, Z. Wu, A. Wang, B. Feng, C. Zou, Y. Sun, N. B. Kovachki, Z. E. Ross, *et al.*, "InverseBench: Benchmarking plug-and-play diffusion priors for inverse problems in physical sciences," in *The Thirteenth International Conference on Learning Representations*.